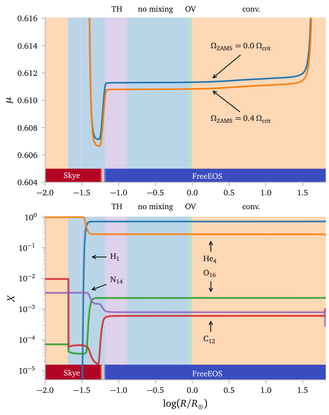
Koen Essers • Master Internship • October 21, 2026

Notes Week 33

a. Thermohaline mixing

Towards the end of the giant branch, we have a mean molecular weight transition between the Skye EOS (Jermyn et al., 2021) and the FreeEOS tables which occurs in a radiative part of the star. This transition causes a *higher* mean molecular weight region to be *above* a lower mean molecular weight region. Even though this mean molecular weight transition is unphysical and entirely caused by the differences in the atom masses used to calculate μ for both tables, *MESA* should detect this transition as a region which is unstable to thermohaline (TH) mixing.

In the plot below, I show the μ profiles, together with the EOS tables and the mixing regions for a star at the tip of the RGB.

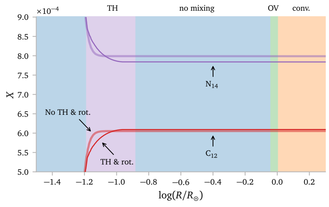


We can quite clearly see that MESA recognises the thermohaline instability and applies mixing in this region. The region is just above the transition from Skye to FreeEOS.

I tested a rotating model too, to see if the thermohaline mixing would be significantly affected by the rotation, however, both models, even though they do experience thermohaline mixing, maintain their μ dip.

This makes a lot of sense if we think about how thermohaline mixing works and how we would usually expect the μ profile to change due to TH mixing. Usually, the instability occurs due to different compositions in different layers. TH mixing then mixes these compositions, which in turn equalized the mean molecular weight profile. However, in this case, the composition is actually nearly identical throughout the mixing region, and even if we completely mix the otherwise unmixed region, the composition stays the same, and we maintain the μ profile dip due to the differences in the opacity tables.

However, for nitrogen and carbon, there are some small differences in the composition in the thermohaline mixing regime. This is a nice test-case to see the effects that "fake" thermohaline mixing can have on the composition of the star.

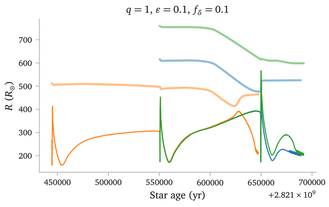


In this case, we can compare the composition of the RGB model with and without thermohaline mixing and rotation. We see that the composition boundary becomes "smoothed" due to the thermohaline mixing, which is what we expect. However, the thermohaline mixing should not even exist here so this effect is actually not physical and wrong.

b. Envelope mass

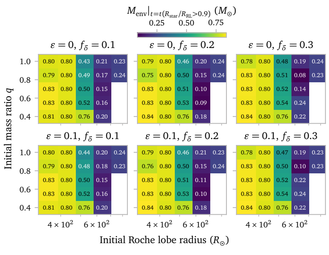
b.1 Introduction

Last week, I showed a couple of plots in which RLOF occured during the interpulse period of the TPAGB star. When the mass ratios were not extreme, and not a lot of mass was lost to a circumbinary disk, RLOF appeared to be very stable[1]. For example, see this figure:



1 In the sense that the radius of the star never became larger than the Roche lobe radius.

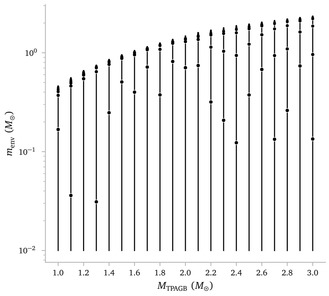
Maybe this surprising stability is due to the low envelope mass at the moment of RLOF. Below, I plot an approximate mass for the start of RLOF. It is approximate because it is difficult to select an exact starting moment. This is also why there are differences in the last two columns.



The third column shows models that interact during the last interpulse phase, where the radius of the star never exceeds the Roche lobe radius. As we can see, the envelope mass at this point is already quite small. But really, to know if this is really the reason, we should compare if with stars with different masses, but equally small evelope masses before the final thermal pulse.

b.2 Setup

Let me plot the masses of the envelope for the different thermal pulses as a function of the mass.

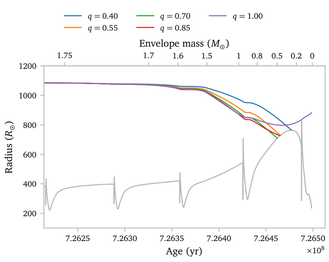


The dots represent the mass of the TPAGB star at the start of a certain thermal pulse. The exact thermal pulse numbers are irrelevant here. It is only important to look at the lowest two dot. These dot represent the mass of the envelope during the *last* two thermal pulse. Clearly, if we interact during the last *interpulse* phase, RLOF will occur when the star has an envelope mass between these two values.

The figure above shows that for the models that interact during the interpulse, the envelope mass is approximately 0.5 $M_\odot$. This figure shows that this is totally in agreement with the values that we expect are possible. The envelope mass of the 1.4 $M_\odot$ star during the last thermal pulse is not very low compared to the other stars.

There are some quite heavy stars that have *lower* envelope masses towards the end of the last interpulse period. For example, it might be interesting to look at the 2.4 $M_\odot$ model. This one has an envelope mass during the last thermal pulse close to 0.1 $M_\odot$, about 2× lower than the 1.4 $M_\odot$ model.

I will use the simple binary code to estimate a nice Roche lobe radius that hits causes RLOF to happen towards the end of the last interpulse, when there is very little mass left. It is then interesting to see if we again obtain these stable RLOF mass transfer binary systems.



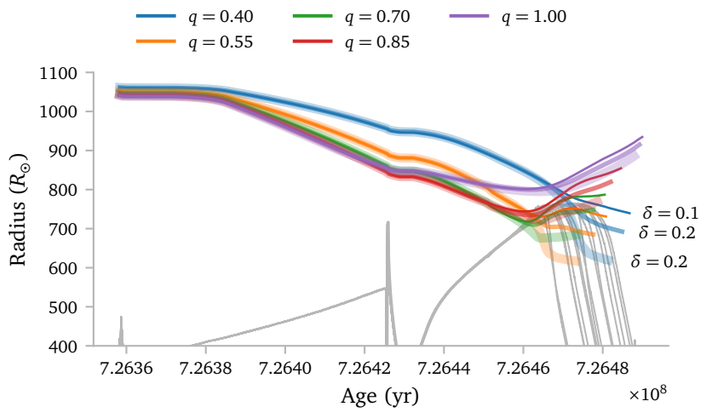
I ended up selecting a single initial Roche lobe radius of $R_{RL} = 1100\ R_\odot$. I simulate the same 5 initial mass ratios as I do for my large grid, and the same 3 f_δ values, however this time only for $\varepsilon = 0$, which means $f_\delta = \delta$.

This seems like it will give interesting results. The models *should* undergo RLOF with very little envelope mass left, or in the case of the $q = 1$ model, experience so sort of flipping where now, wind mass-loss causes an increase in the Roche lobe radius. This is probably some non-trivial effect of the changing mass ratio in the prescription from Saladino et al. (2018) and a nice consistency check. I expect roughly the same behaviour in the detailed *MESA* models.

I expect there to be *big* differences in the final periods *and* the final [s/Fe] values for these models. This is simply because this heavier star reaches much larger radii and much larger envelope s-process abundances.

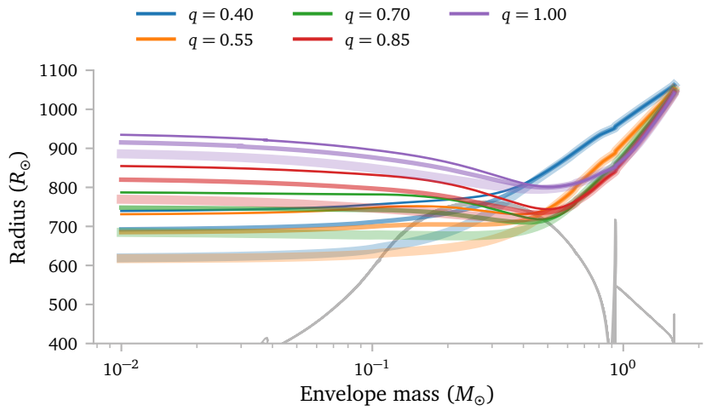
b.3 Results

I simulated the models using *MESA*. Below, I first just plot their Roche lobe evolution as a function of time.

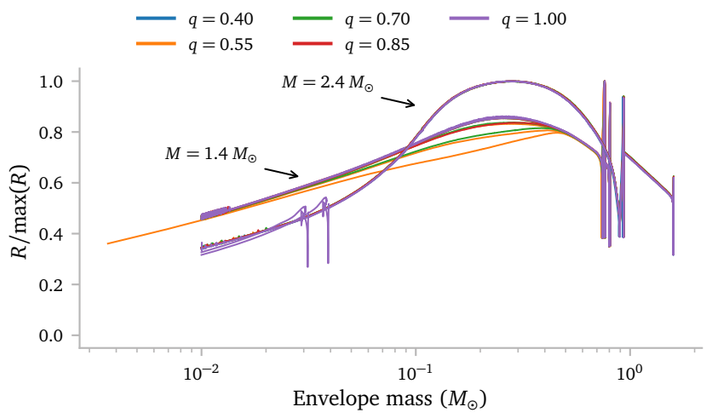


Although it is a bit cluttered, we can definitely see that the detailed *MESA* models have very similar Roche lobe radius evolutions.

We can also see that larger values of δ consistently have lower final Roche lobe radii and shorter lifetimes, thus larger mass loss rates.



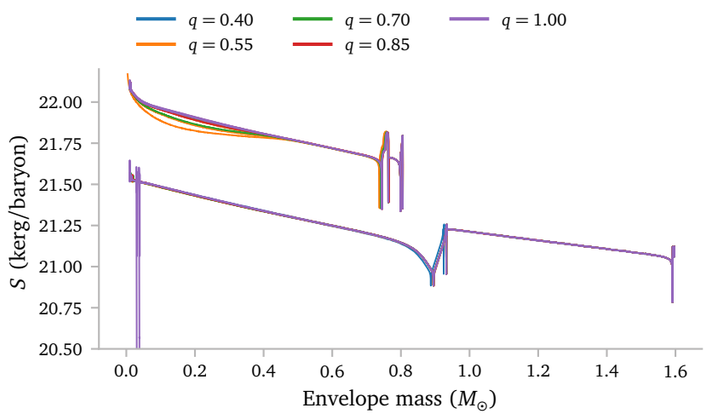
Now, let me compare only the radius of the stars to the low mass models and something should become very obvious:



Clearly, for some reason, the lower mass models *do* seem to respond radially to the mass transfer, generally *shrinking*, while the higher mass star does not, even though they have roughly the same envelope mass remaining when undergoing Roche lobe overflow. The actual radius of the star is however very different.

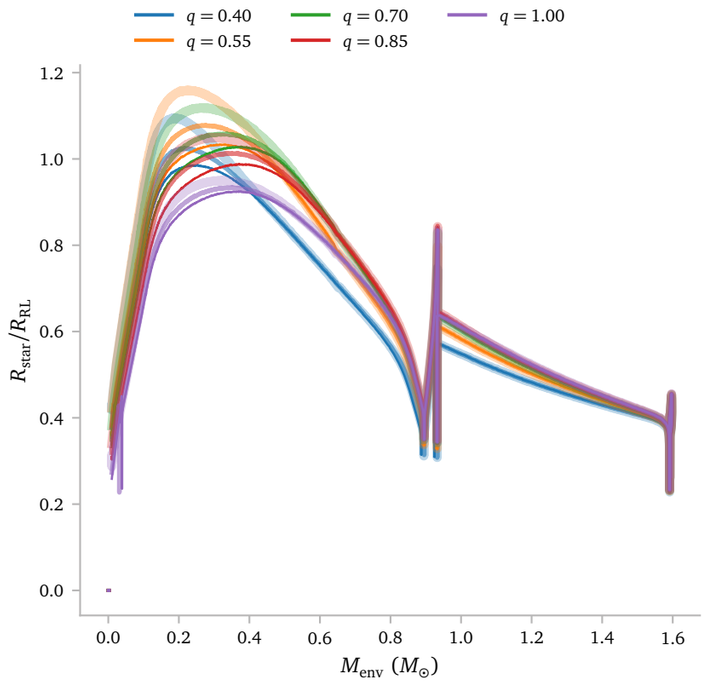
The difference in actual radius is probably mostly determined by the core mass, which is larger for the more massive star, even though the envelope masses are similar when RLOF happens.

It might be interesting to look into the entropy profiles of the stars. Maybe we can see something there that can explain why the lower mass stars are able to decrease their radius as a function of mass loss, while the higher mass star cannot.



First of all, there is a clear difference between the general value of the minimum entropy inside the envelope of the star. The 1.4 $M_\odot$ star has much higher entropy. However, both stars seem to behave quite similar in response to the mass loss. For the 2.4 $M_\odot$ model, the different mechanisms of mass loss really do not affect the entropy profile at all, while for the 1.4 $M_\odot$ model, we see some minor differences, with the most extreme mass transfer mechanisms producing slightly shallower. However, neither show anything interesting, which is good I guess because that means the star is in thermal equilibrium.

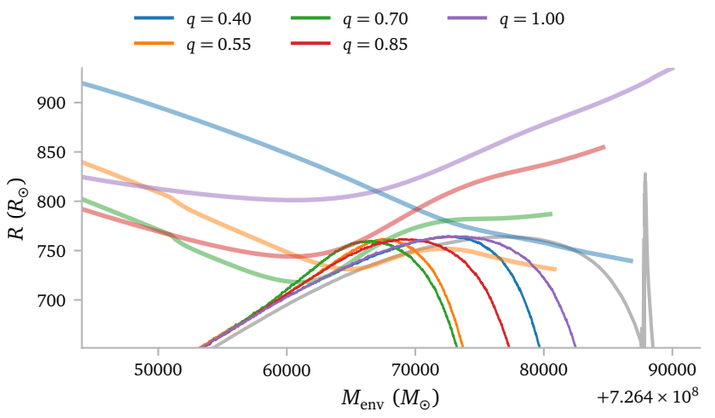
The figure below shows the ratio between the radius of the star and the Roche lobe radius.



Clearly, none of the models reach very large overfill ratios and there are again some models that do not reach overfill ratios larger than 1 even though they do seem to experience semi-RLOF.

Let me actually plot these models separately so they are easier to distinguish.

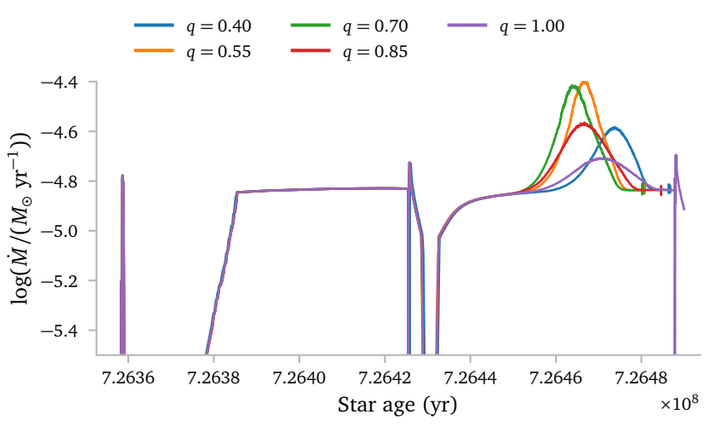
In this case, I show various q, but only the $\delta = 0.1$ models. These have the least extreme RLOF.



Compared to the 1.4 $M_\odot$ models, these are a lot messier in the sense that the Roche lobe radii cross and earliest interacting models have a q somewhere between 1 and 0.4 where for the 1.4 $M_\odot$ models, the $q = 0.4$ models interacted first, then 0.55 up to 1.

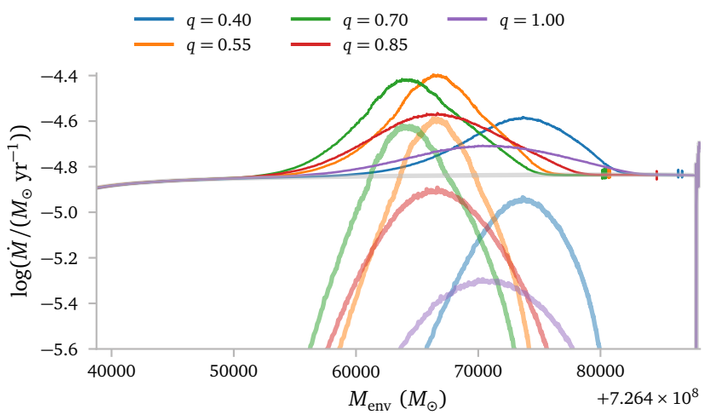
However, we can clearly see that for the blue, red and purple models, the star never has a radius larger than the Roche lobe radius, even though the star loses its mass more quickly here.

Let me plot the mass transfer rates as well.



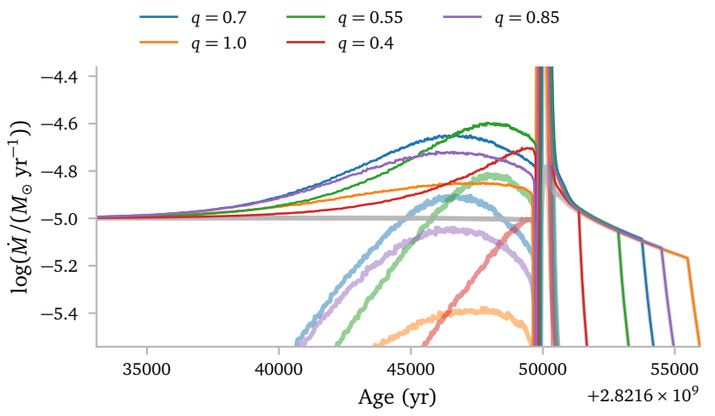
The mass loss rate seems to only be slightly larger then the mass loss rate strictly due to winds.

We can actually split the mass loss in wind and RLOF, but for the sake of clarity, I do that in the figure below.



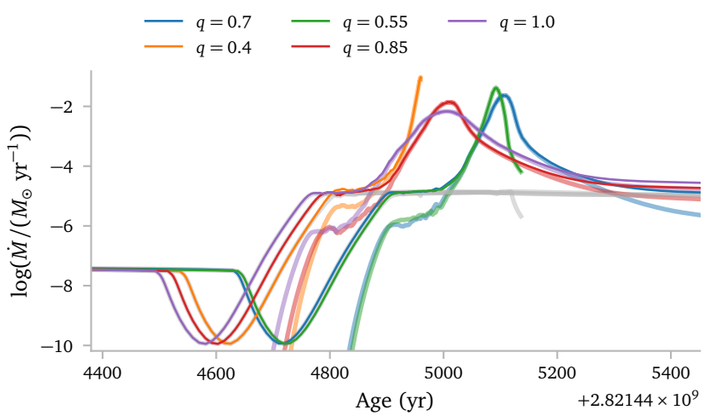
Clearly, for the models without the radius of the star ever exceeding the Roche lobe radius, the RLOF mass transfer rate never exceed the wind mass loss rate.

We see something similar if we look at the 1.4 $M_\odot$ models that interact during the last interpulse.



Again, we see that the RLOF mass transfer rate is comparable to the wind mass loss rate at the RLOF peak.

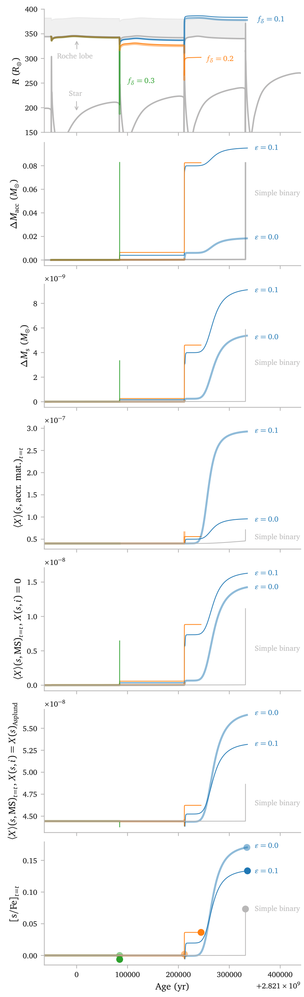
However, if we look at the same 1.4 $M_\odot$ models for a smaller initial Roche lobe radius, we see that the RLOF mass loss rate is much larger. For example, here I plot models with $R_{\rm RL} \approx 427$.



Here, we obtain *much* larger RLOF mass transfer rates than the wind mass loss rate.

c. Mixing mass and the Effect on the final abundances

Last week, I showed this plot:

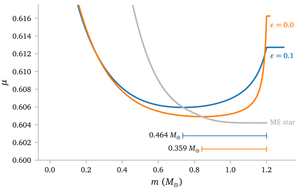


Every plot here makes some intuitive sense if we consider that the envelope s-process abundances increase after thermal pulses up to the point were we start actually *accreting* material onto a main sequence star, and we obtain MS abundances.

This is where things start to become a little difficult to predict. This panel shows the abundances in a MS accretor if the MS itself has no initial s-process abundances. To understand these abundances, we need to consider both the mixing mass, *and* the accreted mass.

For the actual main sequence abundances, it is important to also consider the initial abundances of the s-process elements in the main sequence star.

Let me plot the μ-mixing profiles for the blue models ($f_\delta = 0.1$). That way, we can see how deep the thermohaline mixing should extend, and what the differences are between the two.

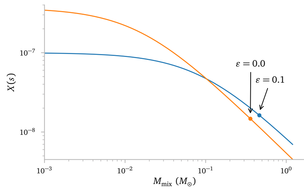


The orange line has less accretion, but with higher s-process abundances, and thus a higher mean molecular weight. The lower accretion mass is more important in this case. The $\epsilon = 0.0$ model has a lower mixing mass.

To confirm the mixing mass actually fixes MS abundances, let me show, explicitly, a calculation of the final accretor abundances. First, let's consider the case *without* any initial abundances in the accretor. Then we simply have

$$X_{\rm MS} = \Delta M_{\rm s\text{-}process}/(M_{\rm mix} + \Delta M_{\rm accr})$$
$$X_{\rm MS,\,\epsilon=0.1} \approx 9 \times 10^{-9}/(0.09 + 0.464) \approx 1.62 \times 10^{-8}$$
$$X_{\rm MS,\,\epsilon=0.0} \approx 5.5 \times 10^{-9}/(0.015 + 0.359) \approx 1.47 \times 10^{-8}$$

These values, although approximate, align very well with what we see in the subplot showing $\langle X\rangle(s, \text{MS})_{t=t}, X(s,i) = 0$



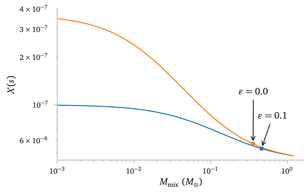
Here I plot the main sequence abundance as a function of the mixing mass. We can clearly see that, in the limit of $M_{\rm mix} \to 0$, $X_{\rm MS}(s) = X_{\rm accr.\ mat.}(s)$. Since the *abundances* of the accreted material in the $\epsilon = 0.0$ case are higher, we end up with larger final abundances in the MS star too. Then, as we go to larger mixing masses, there comes a crossing point where the *quantity* of the s-process elements is just as important as the proportion of s-process elements in the accreted material. Then, towards even higher masses, we reach the actual mixing masses and we can see that, even though the $\epsilon = 0.1$ model has a deeper mixing mass, it still produces a larger final accretor abundance.

Now, if we consider the initial abundances of the accretor, I calculate the abundances as follows:

$$X_{\rm MS} = (\Delta M_{\text{s-process}} + X(s) \cdot M_{\rm mix})/(M_{\rm mix} + \Delta M_{\rm accr})$$
$$X_{\rm MS,\,\epsilon=0.1} \approx (9 \times 10^{-9} + 4.4 \times 10^{-8} \cdot 0.464)/(0.09 + 0.464) \approx 5.31 \times 10^{-8}$$
$$X_{\rm MS,\,\epsilon=0.0} \approx (5.5 \times 10^{-9} + 4.4 \times 10^{-8} \cdot 0.359)/(0.015 + 0.359) \approx 5.69 \times 10^{-8}$$

Both of these values align very well with the plot showing $\langle X\rangle(s, \text{MS})_{t=t}$, $X(s,i) = X(s)_{\rm Asplund}$.

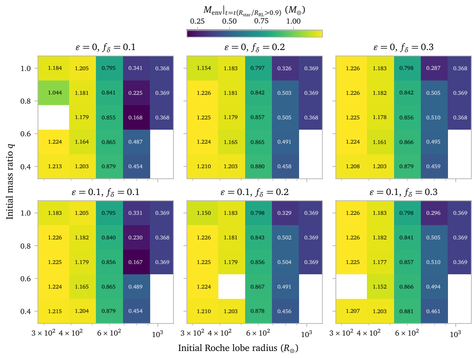
Again, we can plot the effect of the mixing mass.



Again, we see that, in the limit of $M_{\rm mix} \to 0$, $X_{\rm MS}(s) = X_{\rm accr.\ mat.}(s)$. This time however, as we increase the mixing mass, we approach the initial abundance of the main sequence accretor. The lines never cross. The $\epsilon = 0.0$ model has a lower mixing mass, and a larger accretor abunance, even though it accreted less material.

d. 1.8 Solar mass grid

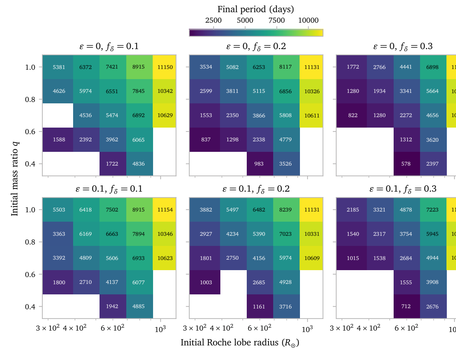
Now, I look into the 1.8 $M_\odot$ grid. Let me start by looking at the mass of the envelope at the approximate start of RLOF.



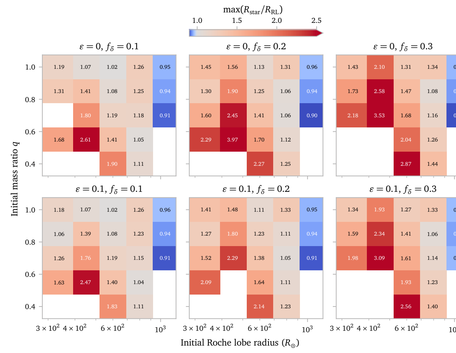
We see quite an important difference between the 1.4 $M_\odot$ models and the 1.8 $M_\odot$ models, with the most important differences being in the third row. Here, there is still a significant amount of mass in the envelope when RLOF occurs. The two models that are consistently missing on the bottom right have not been simulated because they miss RLOF by a lot. The other empty cells are models that were not able to finish their simulation in the maximum 2-day time limit.

There are some models that spiral in and there is one model that fails, even though it does not spiral in. This is the $\varepsilon = 0.1$, $\delta = 0.3$, $q = 0.55$, $R_{\rm RL} \approx 448\ R_\odot$ model.

This model is in an inspiral, but reaches some unstable state during the inspiral. This is definitely not a stable model and it is not problematic that it fails. We can just classify as an inspiral model.

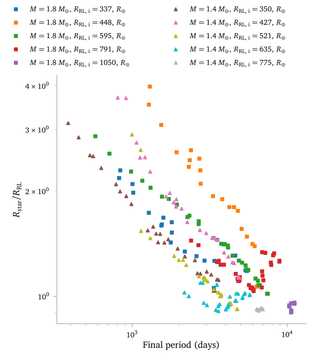


We see quite a large range of possible periods, from 578 to 11154 days. The smallest period here actually occurs in a model with a relatively large initial Roche lobe radius. I *think* this is kind of accidental, and this is really the model that is most close to the unstable inspiral regime. We can also plot the maximum ratio between the radius of the star and the Roche lobe radius. We then see that the models with relatively small final periods also reach larger overfill ratios.



However, this is not the only this that matters. From the two plots above, it seems like models with initially smaller Roche lobe periods reach lower maximum overfill ratios while still producing small final periods. Intuitively this makes sense too.

We can make a scatter plot of all models with the final period versus the maximum overfill ratio.

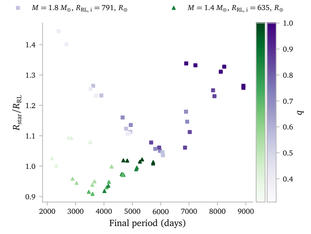


We see quite clear diagonal stripes. They almost look like they would fit a powerlaw nicely. The mass does not seem to affect the slope of the diagonal lines, neither does the initial Roche lobe radius. However, this initial Roche lobe radius quite drastically changes the vertical offset. I think this has to do with *where* in the radius evolution RLOF occurs. For example, if the model interacts towards the top of a thermal pulse, I would expect less overfill for all models compared to models that interact towards the bottom of a thermal pulse.

The purple squares and the gray triangles occupy only a very small region of the plot. This is because these models all miss RLOF and are wind only models.

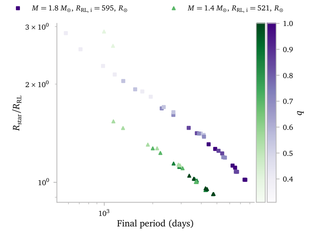
The cyan triangles and the red squares do not follow the nice diagonal line. These are models that interact during the last thermal pulse. Let me try to understand why this is.

Let me first split the models different q.



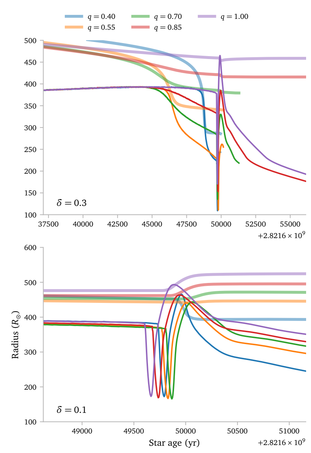
We clearly see that the larger q values are on the upper right hook, while the low q models are in the upper left.

For the normal diagonal lines, splitting per q looks like this:

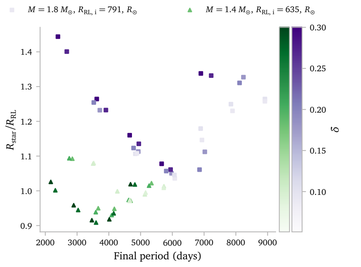


For the normal models, we see that the lower q models are again at the top left with the smallest final periods and the largest overfill ratios. The largest q models have the smallest overfill ratios, but the largest final periods. We however do not see the distinct hook. I really want to understand where the hook comes from.

On another note, the green models show two markers distinctly away from the other markers. These are two models that interact one thermal pulse before the other models do. Contrary to what I expected, these models have *larger* overfill ratios, even though they interact at the *top* of the thermal pulse.



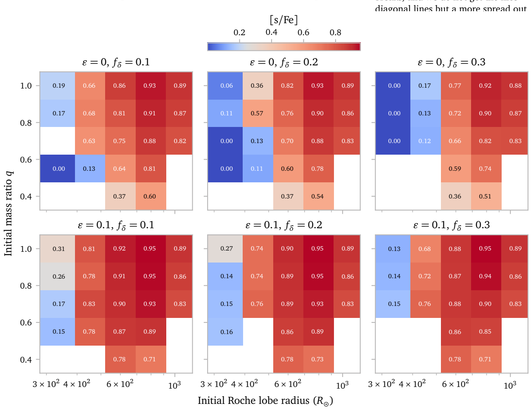
It looks like δ has some effect on where in the radius evolution the models interact. If we have a relatively weak delta of 0.1 the models interact in the thermal pulse with higher final periods and higher overfill ratios. We should be able to see this in the scatter plot too if we color the models based on delta.



This figure, combined with the q, figure and the evolution plots provide quite a nice explanation of the slanted u shape.

Most of the $\varepsilon = 0.3$ and $\delta = 0.2$ models interact during the last interpulse period. Only certain models with even mass ratios still interact during the thermal pulse. This quite naturally explains why we see the downwards diagonal lines in both the 1.4 $M_\odot$ and the 1.8 $M_\odot$. Then, there are some scattered models with large q, (see the figure above) that miss this interaction. We also have the many different q models with $\delta = 0.1$, and certain models with $\delta = 0.2$ that miss the interpulse and interact during the last thermal pulse. Due to some complicated wind effects, the trend breaks, and we do not get the nice diagonal lines but a more spread out

Then we can look at the s process abundances in the form of the 2d grids.

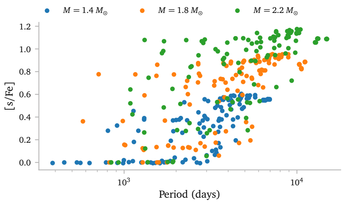


Again, we see that in the case of $\varepsilon = 0$, we have more models with lower barium abundances with lower initial Roche lobe radii. This is because there is nearly no wind in this region.

e. 2.2 Solar mass models

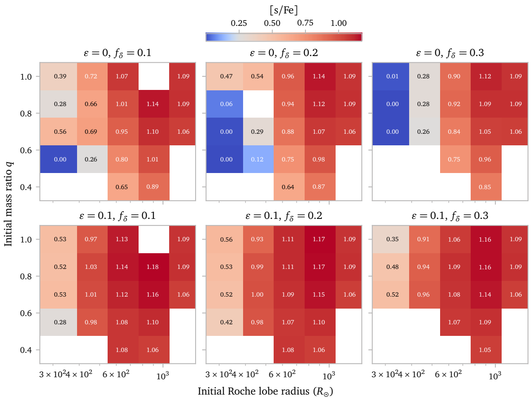
This morning, the 2.2 $M_\odot$ models also finished.

Below, I compare the final periods and the [s/Fe] values for the different masses.

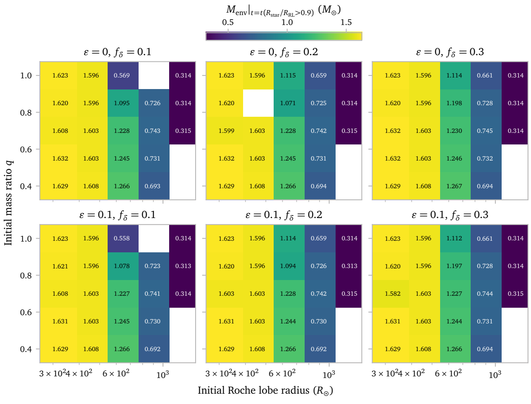


We can quite clearly see that the 1.4 $M_\odot$ models clearly produce the shortest period binaries, but also the least s-process abundant accretors.

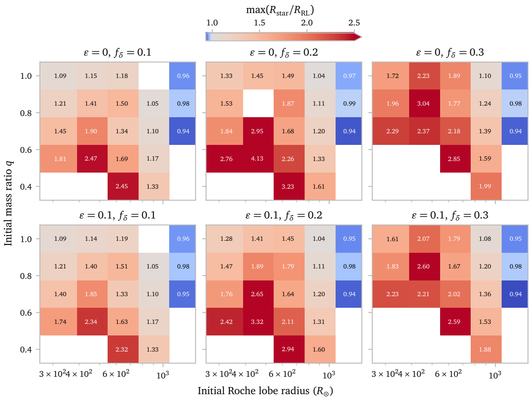
Below I plot the [s/Fe] values.



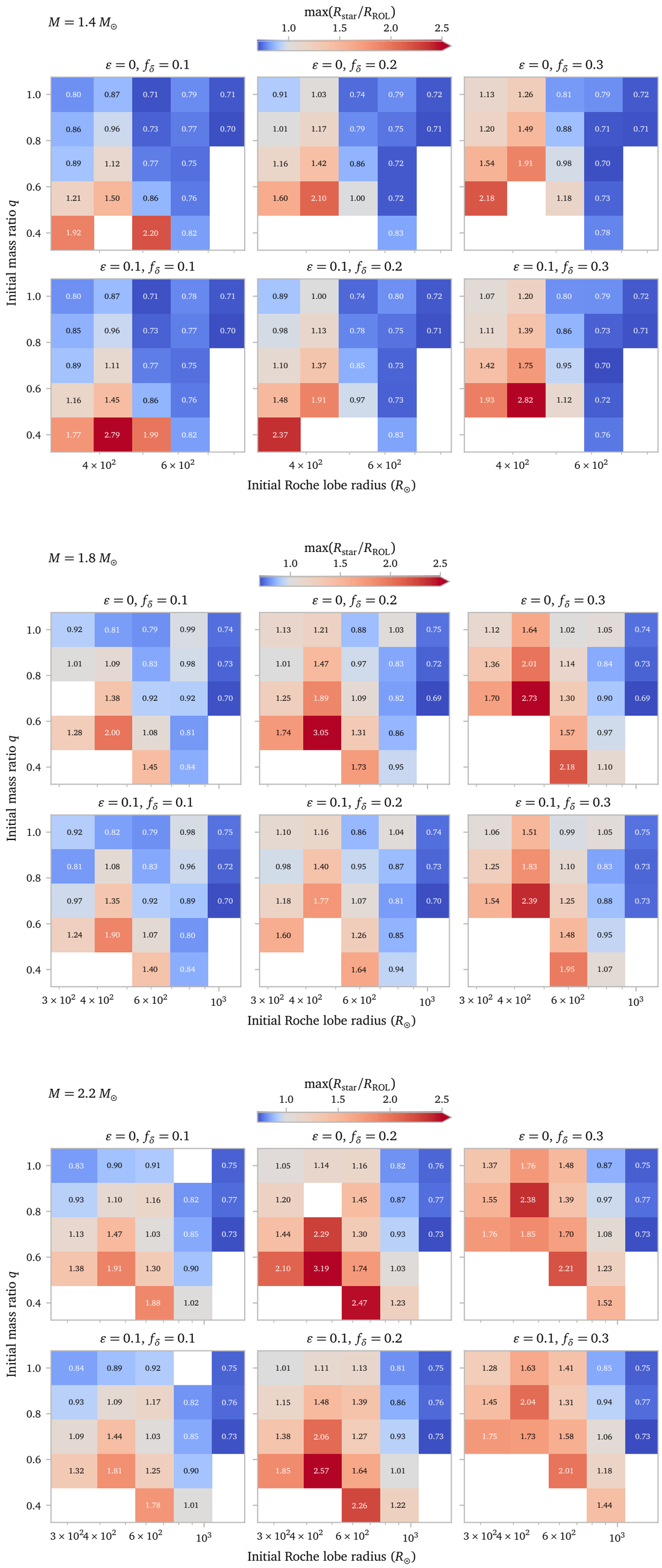
Then I plot the amount of envelope mass left at the approximate start of RLOF.



Lastly, I plot the maximum overfill ratio:



f. Outer Roche lobes



References

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